

ANGRA DO HEROISMO, Portugal

WMO: 085110

Lat: 38.6600N

Lon: 27.2242W

Elev: 298

StdP: 14.54

Time Zone: -1.00 (AZR)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	48.4	50.7	38.5	34.7	51.1	40.3	37.3	52.5	29.4	58.2	26.1	57.5	8.2	310	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	78.3	71.5	77.1	70.7	76.0	70.0	73.9	76.0	72.8	75.3	71.8	74.5	8.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	125.0	75.1	71.9	119.3	74.5	70.8	114.6	73.8	37.6	76.0	36.5	75.5	35.6	74.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	20.5	17.9	DB	44.2	81.3	9.7	1.4	37.2	82.3	31.5	83.1	26.1	83.9	19.0	84.9	(4)
(5)			WB	42.6	74.6	1.8	0.8	41.3	75.2	40.3	75.7	39.3	76.2	38.0	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	58.8	58.1	58.1	58.4	62.0	66.0	70.4	72.4	70.5	66.6	62.3	59.7	(6)
(7)		DBStd	5.87	2.81	2.89	2.79	2.97	2.57	2.80	2.55	2.16	2.61	3.21	3.32	3.25	(7)
(8)		HDD50	1	0	0	0	1	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	1192	193	193	214	198	98	20	0	0	1	20	90	165	(9)
(10)		CDD50	4982	272	228	251	253	371	481	632	694	615	514	370	301	(10)
(11)		CDD65	699	0	0	0	0	5	51	167	229	166	69	11	1	(11)
(12)		CDH74	831	0	0	0	0	0	12	180	453	175	11	0	0	(12)
(13)		CDH80	6	0	0	0	0	0	0	1	4	1	0	0	0	(13)
(14)	Wind	WSAvg	8.1	9.1	9.1	9.6	8.9	7.5	6.9	6.1	6.3	6.9	8.3	8.3	9.6	(14)
(15)	Precipitation	PrecAvg	41.2	4.3	3.8	4.7	3.0	2.8	1.8	1.2	1.7	3.1	4.7	3.9	6.0	(15)
(16)		PrecMax	59.4	7.8	8.9	11.2	8.3	9.9	4.3	3.0	6.4	9.6	13.1	8.6	13.0	(16)
(17)		PrecMin	28.5	0.9	0.6	1.0	0.8	0.2	0.2	0.3	0.1	0.3	1.1	0.6	1.7	(17)
(18)		PrecStd	7.0	1.7	2.2	2.7	2.2	2.2	1.0	0.8	1.3	2.0	2.8	1.9	3.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.7	66.3	65.8	67.5	70.6	75.6	79.6	80.5	79.0	75.5	70.9	67.5	(19)
(20)			MCWB	62.6	61.2	61.5	60.9	65.0	68.1	71.6	72.0	72.2	69.6	66.8	64.5	(20)
(21)		2%	DB	64.6	64.4	64.2	65.2	68.7	73.4	77.5	78.8	77.3	73.6	69.3	66.3	(21)
(22)			MCWB	61.9	60.6	61.0	60.0	63.7	68.0	70.9	71.9	70.7	68.6	65.7	63.6	(22)
(23)		5%	DB	63.7	63.2	63.0	63.8	67.3	71.9	76.0	77.6	76.1	72.4	67.9	65.3	(23)
(24)			MCWB	61.3	59.9	59.6	59.4	62.7	67.1	70.1	71.0	69.9	68.0	64.6	62.7	(24)
(25)		10%	DB	62.9	62.1	62.0	62.6	66.1	70.5	74.7	76.5	74.8	71.1	66.9	64.4	(25)
(26)			MCWB	60.6	59.1	58.8	58.7	61.8	65.9	69.2	70.3	69.3	67.3	63.8	61.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.8	63.8	63.7	63.0	67.1	71.3	75.1	75.6	74.6	71.8	68.8	66.1	(27)
(28)			MCDWB	64.9	64.4	64.4	64.7	68.8	72.7	76.5	77.5	76.4	73.4	69.9	66.6	(28)
(29)		2%	WB	63.0	62.1	62.1	61.8	65.5	69.6	73.5	74.1	73.1	70.2	67.2	64.6	(29)
(30)			MCDWB	63.7	63.2	63.2	63.4	67.3	71.9	75.6	76.5	75.3	72.0	68.2	65.6	(30)
(31)		5%	WB	62.1	61.0	60.9	60.9	64.2	68.3	72.1	73.0	72.0	69.3	66.0	63.6	(31)
(32)			MCDWB	63.1	62.2	62.1	62.4	66.1	70.7	74.6	75.7	74.2	71.2	67.2	64.7	(32)
(33)		10%	WB	61.2	60.1	59.9	60.0	63.0	67.0	70.7	72.0	70.9	68.4	64.7	62.6	(33)
(34)			MCDWB	62.4	61.6	61.5	61.8	65.2	69.5	73.6	75.2	73.5	70.5	66.5	63.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.9	6.7	7.3	7.3	7.8	8.4	8.6	8.1	7.1	6.3	6.2	(35)
(36)			MCDBR	6.0	7.5	6.9	7.9	8.0	8.6	9.1	9.2	8.8	7.7	6.4	6.1	(36)
(37)			MCWBR	6.0	6.6	5.9	6.2	5.7	5.9	6.0	5.9	6.1	5.8	5.7	6.1	(37)
(38)		5% WB	MCDWB	5.5	7.0	6.2	6.8	7.1	7.7	7.9	8.1	7.4	6.8	6.3	5.4	(38)
(39)			MCWBR	6.3	7.1	6.3	6.3	5.7	5.8	6.0	6.1	6.3	6.0	6.6	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.361	0.370	0.400	0.398	0.389	0.389	0.381	0.374	0.373	0.379	0.362	0.363	(40)	
(41)		taud	2.454	2.438	2.343	2.378	2.439	2.472	2.509	2.550	2.552	2.505	2.483	2.427	(41)	
(42)		Ebn at Noon	256	270	272	279	283	282	283	284	278	264	253	245	(42)	
(43)		Edh at Noon	26	30	37	37	36	35	33	31	30	28	25	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	639	911	1271	1621	1894	2003	2038	1935	1533	1053	718	543	(44)	
(45)		RadStd	50	51	90	137	131	140	115	75	79	55	39	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	+1.03	N/A	+0.50	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page